DISCRETE MATHEMATICS · CLASSROOM EDITION

LOGIC BOMB

Defusal Manual · Version 2.0 (Quick-Reference)

> **Expert, read this first.** The Defuser can see the bomb but not this manual. You can see this manual but not the bomb. Only talking, and the mathematics, connect you. Every module below fits in one box. Find the box, do the math, give the order.

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MATH IN THIS MANUAL

Propositions and connectives · truth tables · conditionals and biconditionals · rules of inference · quantifiers and nested quantifiers · sets, set operations, cardinality · functions as sets of ordered pairs.

An original classroom game inspired by cooperative bomb-defusal video games. Not affiliated with any commercial game.

Defusing a Logic Bomb

WIN AND LOSE

- Disarm **every module** before the timer hits 0:00.
- A wrong action is a **strike**. The bomb explodes on its last strike (3 by default). After a strike the timer runs faster.
- A module is disarmed when its **status light** (top-right) turns **green**. Do modules in any order.
- The timer module has no rules. A **needy** module (if present) never disarms: just keep answering it.

THE ONE LAW

> When a box lists numbered rules, check them **top to bottom** and use **only the first rule that is TRUE**.

TWO REMINDERS

- **`P → Q`** is FALSE only when P is TRUE and Q is FALSE.
- **`∀`** about nothing is TRUE. **`∃`** about nothing is FALSE.

READING THE CASING (THE DEFUSER ROTATES THE BOMB TO SEE THE SIDES)

On the casing	What the Defuser reports	What you write down
Serial number sticker	Six letters and digits, e.g. **`KA4T74`**	**`D`** = set of its digits **`{4, 7}`** **`V`** = set of its vowels **`{A}`**
Batteries	How many battery cylinders (an AA holder has 2, a D holder has 1)	**`b`** = number of batteries
Indicators	Small round lights with a 3-letter label; lit (glowing) or not	Is at least one indicator **lit**?
Ports	Plugs such as USB or PARALLEL	Not used in this edition.

Sets ignore repeats: serial **`7A7B1C`** gives **`D = {1, 7}`**. A serial with no vowels gives **`V = ∅`**.

NOTATION

Symbol	Meaning	Symbol	Meaning
`¬P`	not P	**`∀x`**	for every x
`P ∧ Q`	P and Q	**`∃x`**	there is at least one x
`P ∨ Q`	P or Q (or both)	**`∃!x`**	there is **exactly one** x
`P → Q`	if P then Q	**`A ∪ B`**	union: in A or B
`P ↔ Q`	P if and only if Q (same truth value)	**`A ∩ B`**	intersection: in both
`x ∈ A`	x is an element of A	**`A − B`**	difference: in A but not B
`|A|`	cardinality: number of elements	**`A´`**	complement: not in A
`∅`	the empty set { }	**`f(x)`**	the y with (x, y) ∈ f

1 Wires

3 to 6 horizontal wires. Cut exactly one.

> Wires are numbered **from the top** (wire 1 = top). “Last” means the **lowest** one. **R(w)** red · **B(w)** blue · **Y(w)** yellow · **K(w)** black · **W(w)** white. The domain is the wires on the module.

3 WIRES

1.	¬∃w R(w)	**cut wire 2**
2.	∀w (R(w) ∨ B(w))	**cut the last wire**
3.	∃w W(w)	**cut the last white wire**
4.	*otherwise*	**cut wire 1**

4 WIRES

1.	∃!w R(w)	**cut the red wire**
2.	∀w ¬Y(w)	**cut wire 1**
3.	∃w K(w) ∧ b ≥ 2	**cut the last black wire**
4.	*otherwise*	**cut wire 4**

5 WIRES

1.	∃w K(w) ∧ ∃w W(w)	**cut wire 4**
2.	∀w ¬R(w)	**cut wire 1**
3.	∃d ∈ D (d is odd)	**cut the last wire**
4.	*otherwise*	**cut wire 2**

6 WIRES

1.	∀w ¬Y(w)	**cut wire 3**
2.	∃!w W(w)	**cut the white wire**
3.	∃w ∃v (w ≠ v ∧ R(w) ∧ R(v))	**cut the last red wire**
4.	*otherwise*	**cut wire 6**

2 The Button

One big round button with a colored cap and a word on it. A light strip sits to its right.

PROPOSITIONS

p	The button is red.
q	The label says **TRUE**.
r	b ≥ 2 (two or more batteries)
s	At least one indicator is lit.

RULES

1.	p ∧ q	**TAP**
2.	¬p ∧ ¬r	**HOLD**
3.	q ∨ s	**TAP**
4.	*otherwise*	**HOLD**

TAP = press and let go right away.

RELEASING A HELD BUTTON

While held, the strip lights up. Work out the strip’s set. Release when the timer shows a digit from that set **in any position**.

Strip	Release on the digit in…
Blue	{1, 2, 3, 4} ∩ {4, 5, 6}
White	{1, 3, 5} − {3, 5}
Yellow	{5, 6, 7} ∩ {1, 3, 5}
Red	{x ∈ ℤ | 1 < x < 3}

Tip: work out all four sets before the Defuser presses. Once they are holding, they can only tell you the strip color.

3 Set Keypad

Four square keys, each printed with a set.

> Press the keys in order of **cardinality**, from the **smallest** set to the **largest**. No two keys have the same size. A correct key lights green; a wrong key is a strike (correct keys stay lit).

THE SETS

A = {1, 2, 3, 4}
B = {3, 4, 5}
C = {2, 4, 6, 8}
Universe **𝕌 = {1, 2, 3, 4, 5, 6, 7, 8}**, so **A′ = 𝕌 − A**.

Careful: **|∅| = 0** but **|{∅}| = 1** (one element: the empty set). **{A, B}** has 2 elements: the sets A and B.

WORKSHEET

Key shows	Elements	| |	Order

4 Simon Maps

A diamond of four colored pads. It flashes a sequence that grows by one each stage.

> For every flashed color **x**, the Defuser presses **f(x)**, in the same order, for the **whole** sequence shown so far. Which function? **If V ≠ ∅** (serial has a vowel) use **f**. **Otherwise** use **g**.

If V ≠ ∅	**f = { (Red, Blue), (Blue, Red), (Green, Yellow), (Yellow, Green) }**
If V = ∅	**g = { (Red, Yellow), (Blue, Green), (Green, Blue), (Yellow, Red) }**

Read **(Red, Blue) ∈ f** as "**f(Red) = Blue**": flashed red → press blue. The sequence has 3 to 5 stages. A wrong press is a strike and the stage starts over.

5 Who’s on ∀irst

A screen showing a quantified statement and six word buttons. Three stages.

> Domain: **everyone in the class**. `T(x, y)` means “**x is taller than y**”. Translate the screen into English, find that sentence below, and the Defuser presses the button with its **code word**.

English sentence	Code word
Someone is taller than everyone.	**COMET**
Everyone has someone taller than them.	**EMBER**
No one is taller than anyone.	**GLACIER**
Everyone is taller than someone.	**BISHOP**
Someone is taller than someone.	**DYNAMO**
Someone has everyone taller than them.	**FALCON**
Everyone is taller than everyone.	**ATLAS**
Someone is taller than no one.	**HARBOR**

Watch the order. `∀x ∃y` and `∃x ∀y` say different things. Also check which variable comes first inside `T( , )`: the first one is the taller person.

6 Venn Wires

4 to 6 vertical wires. Lights above, panels below.

> Wires are numbered **left to right**. Each wire is red, blue, striped red-and-blue, or white. Each has an LED above it (on/off) and a panel below it (★ or blank).

SETS

R	wires with red on them (red or striped)
B	wires with blue on them (blue or striped)
L	wires whose LED is on
S	wires with a ★ below

> Cut every wire in `K = (R ∩ L) ∪ (S − B)`.

Order doesn’t matter. Cutting a wire **not** in K is a strike. A white wire is in neither R nor B.

WORKSHEET (✓ = YES)

#	R	B	L	S	R∩L	S−B	Cut?
1							
2							
3							
4							
5							
6							

7 Quantifier Grid

A green screen with a 3 × 3 grid of digits, and four round colored buttons.

> Rows **r** go top to bottom, columns **c** go left to right, and **x** is the digit in row r, column c. Press the color of the first TRUE rule.

1.	`∃r ∀c (x is even)`	**RED**
2.	`∀r ∃c (x = 0)`	**BLUE**
3.	`∃c ∀r (x > 5)`	**GREEN**
4.	*otherwise*	**YELLOW**

How to say them:

1. “Is there a **row** where **every** digit is even?”
2. “Does **every row** contain a 0?”
3. “Is there a **column** where every digit is bigger than 5?”

0 counts as even.

8 Truth Table

A screen with a formula in P and Q, four numbered T/F windows with arrows, and a SUBMIT key.

> Set each window to the formula’s truth value in that row of the truth table, then press **SUBMIT**. Either arrow flips T ↔ F. A wrong SUBMIT is a strike.

Window	P	Q	Formula
1	T	T	
2	T	F	
3	F	T	
4	F	F	

P	Q	¬P	P∧Q	P∨Q	P→Q	P↔Q
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Tip: ¬ binds tighter than everything else: `¬P ∨ Q` means `(¬P) ∨ Q`, but `¬(P ∨ Q)` negates the whole thing.

N Needy: Argument Check

A screen asks “VALID ARGUMENT?” with Y and N keys and a 2-digit countdown. It wakes up every so often.

> When it shows an argument (premises, a line, then ∴ conclusion), press **Y** if it is **valid** and **N** if it is **invalid**, before the 40-second countdown ends. Wrong answer or time out = strike. Match the pattern:

VALID → press Y	
`p → q, p ∴ q`	modus ponens
`p → q, ¬q ∴ ¬p`	modus tollens
`p → q, q → r ∴ p → r`	hypothetical syllogism
`p ∨ q, ¬p ∴ q`	disjunctive syllogism
`p ∴ p ∨ q`	addition
`p ∧ q ∴ p`	simplification
`∀x P(x) ∴ P(c)`	universal instantiation
`P(c) ∴ ∃x P(x)`	existential generalization
`∀x (P(x) → Q(x)), P(c) ∴ Q(c)`	universal modus ponens
`¬∀x P(x) ∴ ∃x ¬P(x)`	negating ∀

INVALID → press N	
`p → q, q ∴ p`	affirming the conclusion
`p → q, ¬p ∴ ¬q`	denying the hypothesis
`p ∨ q, p ∴ ¬q`	“or” is not “exclusive or”
`∀x (P(x) → Q(x)), Q(c) ∴ P(c)`	affirming the conclusion (∀)
`∃x P(x) ∴ ∀x P(x)`	hasty generalization
`¬∀x P(x) ∴ ∀x ¬P(x)`	bad negation

B Saying symbols aloud

Agree on these words before the timer starts.

Symbol	Say	Symbol	Say	Symbol	Say
∀	“for all”	∪	“union” / “cup”	∅	“empty set”
∃	“there exists”	∩	“intersect” / “cap”	{ }	“set of …”
¬	“not”	−	“minus”	′	“prime” (complement)
∧	“and” / “wedge”	→	“arrow” / “implies”	∴	“therefore”
∨	“or” / “vee”	↔	“double arrow” / “iff”	★	“star”

Worked examples

Casing for every example: serial **KA4T74** → **D = {4, 7}**, **V = {A}**; one AA holder → **b = 2**; indicator NEG lit.

Module	Bomb shows	Work	Answer
1 · Wires	4 wires: blue, red, blue, yellow	Exactly one red, so rule 1 **∃!w R(w)** is TRUE.	Cut wire 2
1 · Wires	5 wires: red, yellow, blue, black, blue	Rule 1 needs a white: FALSE. Rule 2: there is a red, FALSE. Rule 3: 7 ∈ D is odd, TRUE.	Cut wire 5
2 · Button	Blue button, label “NOT”	p F, q F, r T (b = 2), s T. Rule 1: F. Rule 2: **¬p ∧ ¬r** = T ∧ F = F. Rule 3: **q ∨ s** = T.	TAP
2 · Button	Held; strip is yellow	**{5,6,7} ∩ {1,3,5} = {5}**	Release on a 5
3 · Keypad	A ∪ B · ∅ · C − B · B ∩ C	**|A ∪ B| = |{1,2,3,4,5}| = 5**, **|∅| = 0**, **|C − B| = |{2,6,8}| = 3**, **|B ∩ C| = |{4}| = 1**	∅, B∩C, C−B, A∪B
4 · Simon	Flashes red, then red, green	**V ≠ ∅** → use f. f(red) = blue, f(green) = yellow.	blue; then blue, yellow
5 · ∀irst	**∀x ∃y T(x,y)**	“For every x there is a y that x is taller than” = Everyone is taller than someone.	BISHOP
5 · ∀irst	**∃y ∀x T(x,y)**	“There is a y such that every x is taller than y” = Someone has everyone taller than them.	FALCON
6 · Venn	1 red/LED on · 2 blue/LED on/★ · 3 white/★ · 4 striped/LED on · 5 red/★ · 6 blue	**R ∩ L = {1, 4}**, **S − B = {3, 5}** (2 has blue). **K = {1, 3, 4, 5}**.	Cut 1, 3, 4, 5
7 · Grid	7 1 3 / 8 2 1 / 9 5 1	No all-even row. Row 1 has no 0. Column 1 = 7, 8, 9, all > 5: rule 3 TRUE.	GREEN
8 · Truth	**¬P → Q**	Row TT: F→T = T. TF: F→F = T. FT: T→T = T. FF: T→F = F.	T T T F, SUBMIT
N · Argument	**∀x (P(x) → Q(x)), Q(c) ∴ P(c)**	Affirming the conclusion (∀). Q(c) could be true for another reason.	N

WHY THESE ARE THE HARD PARTS

- **Vacuous truth.** Wires rule **∀w ¬Y(w)** is TRUE when there is no yellow wire at all.
- **Quantifier order.** **∀x ∃y** (everyone has a partner) is weaker than **∃y ∀x** (one person works for everyone).
- **Sets of sets.** **{∅}** is not empty. It has one element.